Cognitive Load Due to Market Unit Mismatch
How normalization returns an easier shopping experience

Issue #1: Inconsistent price formats present unnecessary cognitive load on consumers

What problem existed?

The unit price for the same grocery item can be presented using different units or package quantities across retailers.
For example, one store may display a price per pound, another per ounce, and another per package.

Although these formats are often legally compliant, this variability makes identifying the best value unnecessarily difficult for consumers. Even when all stores operate within regulatory guidelines, users must perform manual unit conversions to make meaningful comparisons.

The difficulty is not rooted in mathematical complexity. The difficulty arises from **repetition**: these conversions must be performed repeatedly across many items during a single shopping session.

This repeated effort compounds cognitive load and contributes to decision fatigue during routine grocery shopping.

What existing solutions failed?

Existing regulatory and commercial solutions address pricing compliance, not comparison usability.

Pricing statutes typically require stores to list unit prices using “reasonable” units. However, retailers are free to choose *which* reasonable unit they display. As a result, two fully compliant stores may present the same product using different unit bases, requiring consumers to mentally normalize prices before comparison.

This means that while the information is technically present, the burden of interpretation is placed entirely on the user.

The problem is not that consumers cannot perform these conversions.
The problem is that they are asked to perform them repeatedly, across dozens of items, every week.

No existing solution eliminates this repetition.

What constraints shaped the design of the solution?

The domain of grocery pricing is constrained in several important ways:

- There is a limited, well-defined set of unit types that appear in retail grocery pricing.

- Each unit type has a known and stable conversion formula.
- The cognitive burden does not come from complexity, but from repeated application of these same conversions.

These constraints make the problem well-suited to normalization at ingestion rather than repeated mental conversion at use time.

The unnecessary cognitive load comes from repeatedly performing the same invariant transformations during routine shopping.

What tradeoffs were chosen, and why?

For a large, long-lived data model, workflow maintainability was prioritized over maximum theoretical accuracy.

As long as key invariants remain stable, carefully chosen shortcuts can improve long-term usability without materially affecting decision quality. The goal was not to model every possible edge case, but to remove repeated cognitive effort from routine decision-making.

This tradeoff enables sustained use and reduces user fatigue over time.

Issue #2: Why normalization beats raw display

A shift from maximum transparency to operational readiness

What problem existed?

Even if all pricing data were presented “correctly,” raw data still makes a poor interface for planning and use.

If every store displayed information identically, the workload would not disappear — it would simply be moved downstream. The user would still be responsible for translating that data into planning, allocation, and consumption decisions.

Raw display optimizes transparency, but it does not optimize for operational use.

What existing solutions failed?

Existing consumer tools each provide only a partial view of the overall problem.

- Retail grocery sites support purchasing, but assume that planning occurs outside the system.

- Meal planning and nutrition apps support consumption goals, but operate independently of how food is purchased.
- Budgeting tools track spending, but do not account for consumption patterns or nutritional value.

Each tool requires the user to understand how to translate information between systems. This creates duplicated effort, increases time cost, and compounds cognitive load.

No existing solution captures the full workflow from purchase through planning and use.

This project explores whether consolidating these fragmented tools into a single system can reduce repetitive cognitive load while preserving user agency.

What tradeoffs were chosen, and why?

While raw data maximizes transparency, it requires users to repeatedly perform the same calculations across multiple contexts.

The system instead performs invariant transformations once at data ingestion. By normalizing data into a planning-ready form, the system eliminates repeated mental work while still preserving access to original values when needed.

This represents a deliberate shift from raw transparency toward operational readiness.

What constraints shaped the development of the solution?

The solution is shaped by the need to translate between two fundamentally different systems:

- **Commerce-facing website units**, optimized for retail display
- **Planning-facing meal usage units**, optimized for consumption and scheduling

Because commercial units and meal planning units are inherently separate, only a small set of invariants exist between them. This separation allows transformative freedom without loss of meaning.

As a result, the system prepares data in a planning-ready form and leverages one-time AI assistance to adapt that data to personalized requirements, presenting a merged view that satisfies both purchase and planning needs.